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Forecasting stock return distributions around the globe with quantile neural networks☆

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ABSTRACT

We propose a novel machine learning approach for forecasting the distribution of stock returns using a rich set of firm-level and market predictors. Our method combines a two-stage quantile neural network with spline interpolation to construct smooth, flexible cumulative distribution functions without relying on restrictive parametric assumptions. This allows for accurate modelling of non-Gaussian features such as fat tails and asymmetries. Furthermore, we show how to derive other statistics from the forecasted return distribution, such as the mean, variance, skewness, and kurtosis. The derived mean and variance forecasts offer significantly improved out-of-sample performance compared to standard models. We demonstrate the robustness of the method in U.S. and international markets.

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1. Introduction

Predicting the distribution of stock returns has become increasingly important because modern financial decision making requires more than just point estimates. Traditional approaches that focus on predicting the mean or variance of returns impose restrictive assumptions, such as multivariate normality or quadratic utility, that rarely hold in practice. These methods fail to capture the complexity of investor preferences, which depend on higher-order moments and distributional properties such as asymmetry, tail behaviour, and discontinuities (Gul, 1991; Kahneman & Tversky, 1979; Kimball, 1993; Manski, 1988), as well as features of the data such as non-Gaussianity and heavy tails. To address these limitations, we develop a machine learning-based framework for forecasting the distribution of stock returns using a rich set of firm-level and market variables.

Specifically, this paper presents a novel approach that uses a two-stage quantile neural network combined with spline interpolation to generate predictive distributions. Our method flexibly models non-Gaussian, heavy-tailed return distributions without relying on parametric assumptions. It captures non-linear interactions among a large number of predictors, allows for estimations of key distributional moments, and provides accurate forecasts in high-dimensional settings with low signal-to-noise ratios. Our approach performs well in both U.S. and international markets and delivers significant improvements in out-of-sample loss metrics compared to existing parametric and machine learning benchmarks in the prediction of next-month returns.

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The ability to forecast next-month return distributions rather than isolated moments has a number of key benefits. It supports decision frameworks that require knowledge of the tail risk or asymmetry, such as those used in portfolio optimisation, risk management, and asset pricing. It improves the estimation of standard moments (mean, variance, skewness, kurtosis) through numerical integration, which we show improves predictive performance over direct moment forecasting. Finally, it also enables new empirical investigations into the pricing of distributional characteristics across assets.

We contribute to the literature in two main ways. First, we propose a new method for distribution forecasting that combines quantile neural networks with cubic B-spline interpolation to construct smooth and flexible cumulative distribution functions. Unlike existing approaches, our method is non-parametric, data-driven, and generalises well across the globe.

Second, we show how distributional forecasts can be used to improve asset pricing applications. Specifically, we examine the pricing of predicted quantiles and moments in the cross-section of stock returns. We show that the mean return derived from the interpolated distribution significantly outperforms direct mean forecasts in out-of-sample portfolio returns. Moreover, we find that only the central quantiles are systematically priced, with the profitability of long-short portfolios declining as one moves towards the tails. Contrary to some previous findings, we find no consistent pricing of volatility, skewness, or kurtosis using our distribution-derived estimates. These results provide new insights into the role of higher-order moments in asset pricing and help reconcile previous mixed evidence in the literature (Ang, Hodrick, Xing, & Zhang, 2006; Bali et al., 2016; Bollerslev et al., 2020).

Our work builds on and extends a growing body of research applying machine learning to asset pricing and return forecasting. Recent studies have demonstrated the superiority of machine learning methods in capturing complex non-linear patterns and improving out-of-sample prediction accuracy; see e.g. Bryzgalova et al. (2020), Dong et al. (2022), Gu et al. (2020, 2021), Tobek and Hronec (2021). However, most of these studies focus on predicting mean returns or linear factor models. In contrast, our approach targets the entire distribution and goes beyond previous quantile prediction methods by interpolating across a dense grid of quantiles to construct smooth distributions.

We also complement recent contributions to distributional forecasting in asset pricing. For example, Liu (2023) demonstrates the usefulness of quantile regression for forecasting returns and constructing quantile-based risk premia. Our approach differs in that we interpolate a richer set of quantile forecasts onto a denser representation of the distribution and derive distributional moments via numerical integration, rather than relying on pre-specified interquantile ranges. We also show how to adjust the predicted higher moments to account for the lack of extreme observations in the tails of the distribution. Furthermore, our method generalises to international markets and is validated using models trained only on U.S. data.

Another related strand is the work of Yang et al. (2024), who use a conditional quantile variational autoencoder (CQVAE) to approximate future return distributions. While both approaches use machine learning to generate distribution forecasts, our method uses observable firm-level predictors and explicit quantile targets, in contrast to the latent factor representations used in the CQVAE. In addition, we provide a detailed decomposition of distributional moments and their pricing implications for both U.S. and global equity markets. In contrast, Yang et al. (2024) construct quantile forecasts with a primary focus on mean estimation.

Our contribution is also relevant to the literature on distributional properties in asset pricing (Alexiou & Rompolis, 2022; An et al., 2014; Bessembinder, 2018; Hou et al., 2020; Huang & Li, 2019; Kim & White, 2004). Existing evidence on the pricing of volatility, skewness, and kurtosis is mixed and often relies on ex post sample estimates or option-implied measures, which have limited availability. Our model provides an alternative by using out-of-sample distribution forecasts based on observable predictors to study the pricing of distributional characteristics. We show that only the mean and certain central quantiles carry significant pricing information, while higher moments do not appear to be systematically priced.

In summary, this paper presents a novel machine learning-based method for forecasting stock return distributions and provides empirical evidence of its predictive accuracy and relevance for asset pricing. Our results underscore the value of distribution forecasting in financial applications and suggest that incorporating richer distributional information can improve return forecasting and portfolio performance.

The rest of the paper is organised as follows: Section 2 presents the distributional forecasting methodology, including quantile neural networks and spline interpolation. Section 3 describes the empirical setup and data, and reports the results of out-of-sample forecasting performance and asset pricing applications. Section 4 concludes.

2. Two-stage quantile neural networks

Let the returns $r_{i,t+1}$ collected over $t = 1, \dots, T$ months and $i = 1, \dots, N_t$ individual stocks be random variables with a conditional cumulative distribution function (CDF) $F_{i,t}(r|\mathcal{F}_t)$ for each asset i at each time t.[2] Let $\tau_1, \dots, \tau_K \in (0, 1)$ denote the quantile levels of interest, and define the set $\mathcal{T} = \{\tau_1, \dots, \tau_K\}$. We are interested in forecasting the corresponding conditional quantiles $Q^{\mathcal{T}}_{r_{i,t+1}} = \left\{Q_{r_{i,t+1}}(\tau_1|\mathcal{F}_t), \dots, Q_{r_{i,t+1}}(\tau_K|\mathcal{F}_t)\right\}$. Each quantile is defined as

$$Q_{r_{i,t+1}}(\tau|\mathcal{F}_t) = \inf\left\{r \in \mathbb{R} : F_{i,t}(r|\mathcal{F}_t) \geq \tau\right\},$$

which represents the τ-quantile of the return distribution at time $t+1$ conditional on the information set $\mathcal{F}_t$ available at time t. Here, $F_{i,t}(\cdot|\mathcal{F}_t)$ denotes the asset-specific

[2] In the case of the forecast distribution of stock returns, $Q_{r_{i,t}}(0) = -1$ and $Q_{r_{i,t}}(1) = \infty$.

and time-varying conditional CDF of $r_{i,t+1}$ given $\mathcal{F}_t$. To forecast these distributions, we exploit the panel structure of the data by pooling information across both assets and time periods, rather than relying solely on single-stock historical time series. This panel-data approach enables us to leverage cross-sectional variation in firm characteristics and time-series variation in market conditions simultaneously—a strategy that has proven effective in forecasting the mean of stock return distributions (Gu et al., 2020) and modelling volatility (Bollerslev et al., 2018). By training a single model on the pooled panel, we effectively impose parameter sharing across assets while allowing the predictions to vary with asset-specific characteristics and time-varying market conditions.

To implement this panel-based forecasting strategy, we jointly predict multiple τ-quantiles using a multi-output feed-forward neural network model that aggregates individual τ-quantile losses into a single loss function, as described below in this section. In addition to being computationally more efficient than training individual models for each τ, joint training for all τ-quantiles allows the model to capture the relationship across the entire distribution. As a side benefit, even without introducing penalisation for quantile crossing,[3] the model naturally captures the monotonicity requirement of quantile predictions, since there are no quantiles that are crossed in our quantile forecasts.

2.1. Network architecture

While neural networks have been successfully used in stock return forecasting (Gu et al., 2020), quantile forecasting is challenging because we need to capture both cross-sectional variation across stocks and market-wide variation over time. One way to address this challenge is to use a cross-sectional standardisation of returns to remove the time-series noise. This is possible from an asset pricing perspective, where the goal is to rank individual stocks cross-sectionally to create long-short portfolios. However, since our goal is to forecast quantiles of future raw (non-standardised) stock returns, we propose a two-stage neural network architecture that exploits the predictability of standardised returns in the first stage and generates quantile forecasts of true (non-standardised) stock returns in the second stage. In this way, we allow the neural networks to exploit the information in standardised returns and to adjust for expected market volatility.

More specifically, a set of quantiles of stock returns is modelled using a large set of variables, including stock-level characteristics (e.g. size, value, or momentum) stored in a vector $x_{i,t}$, as well as market variables (e.g. market volatility) stored in a vector z_t, both of which proxy the information available at period t. We use both raw and standardised returns to address the discussion above, and we standardise the return of a stock by dividing the raw stock returns by the cross-sectional average of the stock-level volatilities at period t, such that

$$\widetilde{r}_{i,t+1} = \frac{r_{i,t+1}}{\overline{\sigma}_t}, \tag{1}$$

where $\overline{\sigma}_t = \frac{1}{N_t}\sum_{i=1}^{N_t}\sigma_{i,t}$ is a cross-sectional average of stock-level volatilities, $\sigma_{i,t}$ is a stock-level volatility estimate of stock i at period t, and N_t is the number of stocks in the panel at period t. Ideally, we would use $\overline{\sigma}_{t+1}$ instead of $\overline{\sigma}_t$ to rescale the standardised quantile forecasts back to the raw quantile forecasts of stock returns, but this is not known at period t when the forecast is made. Therefore, we use the cross-sectional average of stock-level volatilities at period t instead of the future value of $\overline{\sigma}_{t+1}$ to avoid look-ahead bias. $\sigma_{i,t}$ is estimated using an exponentially weighted moving average of the squared returns of stock i with a decay factor of 0.94 and is one of the input variables to the model.[4] See Appendix J for a visualisation of $\overline{\sigma}$ over time in the U.S., and more detailed arguments for the two-stage setup.

The idea behind standardising individual stock returns using the cross-sectional average of stock-level volatilities rather than the stock volatility itself is to reduce the noise from individual stock volatility estimates. For the same noise-reducing reason, we standardise stock returns only by dividing by the average volatility and not by subtracting the average return, either stock or market, because average returns are not persistent.

Let $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$ be a set of quantile levels with $\tau_k \in (0, 1)$. A stock return quantile is then modelled by the two-stage quantile neural network with two sub-networks as (see the equation in Box I).

where $W_S = \left(W_S^{(1)}, \ldots, W_S^{(L)}\right)$ and $b_S = \left(b_S^{(1)}, \ldots, b_S^{(L)}\right)$ are weight matrices and bias vectors, respectively, from a stage S. Each weight matrix $W_S^{(\ell)} \in \mathbb{R}^{m\times n}$ contains m neurons as n column vectors $W_S^{(\ell)} = [w_{\cdot,1}^{(\ell)}, \ldots, w_{\cdot,n}^{(\ell)}]$, and $b_S^{(\ell)}$ are thresholds or activation levels that contribute to the output of a hidden layer that allows the function to be shifted.

It is important to note that, in contrast to the literature, we consider a multi-output (deep) neural network to characterise the collection of quantiles. Our network has two sub-networks, namely a standardised τ-quantile network and a market-wide volatility network,[5] in which $l \in 1, \ldots, L$ hidden layers transform the input data into a chain using a collection of nonlinear activation functions $g_S^{(1)}, \ldots, g_S^{(L)}$. An activation function, $g^{(\ell)}_{W_S^{(\ell)}, b_S^{(\ell)}}$, is used as a

$$g^{(\ell)}_{W_S^{(\ell)}, b_S^{(\ell)}}(u) := g_\ell\left(W_S^{(\ell)}u + b_S^{(\ell)}\right) = g_\ell\left(\sum_{i=1}^{m} W_{i,S}^{(\ell)}u + b_{i,S}^{(\ell)}\right).$$

Fig. 1 details the architecture of the two-stage quantile neural network model defined above. This is our main model used for quantile forecasting, and we refer to it as the *two-stage model* throughout the paper. The first stage of the model is used to forecast quantiles of cross-sectionally standardised next-month stock returns $Q^{\mathcal{T}}_{\widetilde{r}_{i,t}}$.

[3] Quantile crossing refers to the situation where $Q(\tau_l) > Q(\tau_h)$ when $\tau_l < \tau_h$, violating the quantile monotonicity condition.

[4] The 0.94 parameter is widely used in both academia and the financial industry, an example being the RiskMetrics GARCH specification.

[5] Note that the market-wide volatility sub-network is forecasting optimal multiplicative adjustment factor to $\overline{\sigma}_t$ to produce quantile predictions for non-standardised stock returns. It does not forecast market volatility directly.

$$
\begin{aligned}
Q^{\mathcal{T}}_{r_{i,t+1}} &= \left\{ Q_{r_{i,t+1}}(\tau_1|x_{i,t}, z_t, \overline{\sigma}_t), \ldots, Q_{r_{i,t+1}}(\tau_K|x_{i,t}, z_t, \overline{\sigma}_t) \right\} \\
&= \Bigg[\underbrace{\left(g^{(L)}_{W^{(L)}_{S_1}, b^{(L)}_{S_1}} \circ \ldots \circ g^{(1)}_{W^{(1)}_{S_1}, b^{(1)}_{S_1}} (x_t) \right)}_{\text{standardised } \tau\text{-quantile sub-network}} \times \overline{\sigma}_t \Bigg] \times \Bigg(\underbrace{g^{(2)}_{W^{(2)}_{S_2}, b^{(2)}_{S_2}} \circ g^{(1)}_{W^{(1)}_{S_2}, b^{(1)}_{S_2}} (z_t)}_{\text{market-wide volatility sub-network}} \Bigg) \\
&= \underbrace{\Bigg[\underbrace{\left\{ Q_{\widetilde{r}_{i,t+1}}(\tau_1|x_{i,t}), \ldots, Q_{\widetilde{r}_{i,t+1}}(\tau_K|x_{i,t}) \right\}}_{\text{Stage I}} \times \overline{\sigma}_t \Bigg] \times \widehat{\sigma}^M_t}_{\text{Stage II}}, \qquad (2)
\end{aligned}
$$

Box I.

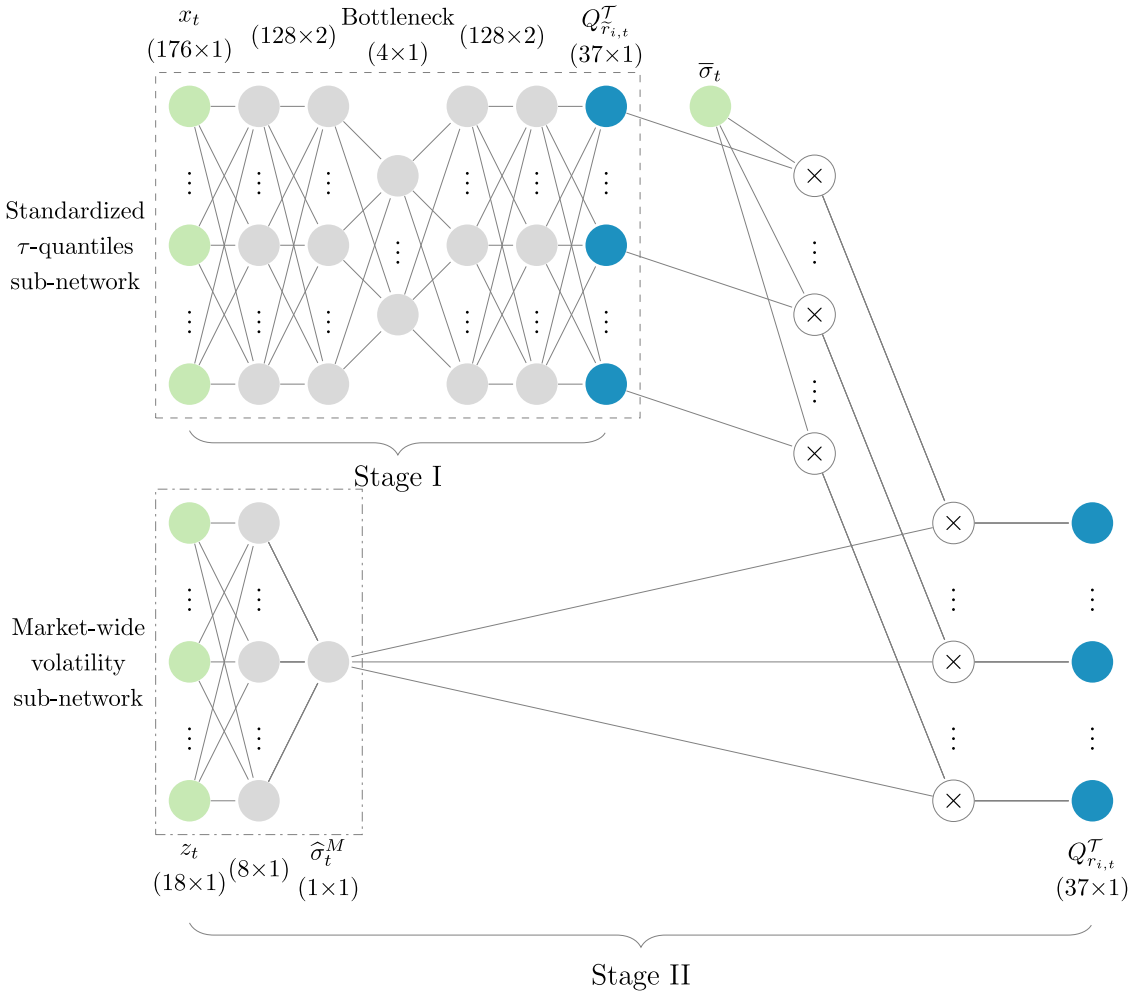


Fig. 1. Two-stage neural network architecture. Stage I generates cross-sectionally standardised τ-quantile forecasts, $Q^{\mathcal{T}}_t$, as output. Cross-sectional standardisation is performed by dividing individual future stock returns $r_{i,t+1}$ (labels) by the cross-sectional average of the individual stock standard deviation estimates, $\overline{\sigma}_t$. The standard deviation estimates for the individual stocks are an exponential weighted moving average of their daily squared returns with a smoothing factor of 0.94. Step II uses $\overline{\sigma}_t$, i.e. based on data from the previous period, to rescale the standardised τ-quantile forecasts generated in Stage I back to the original scale. The rescaling takes place at the multiplication nodes marked with $\times$, representing the multiplication operation between the inputs of the node. The market-wide volatility sub-network then generates a forward-looking scaling factor to $\overline{\sigma}_t$ in order to further refine the τ-quantile forecasts produced as the output of Stage II. The forward pass generates a tensor of form $(B, 37, 2)$, where B is the number of observations in the batch, 37 is the number of predicted quantiles, and 2 corresponds to the standardised and raw versions.

In our empirical application, we use 37 τs ranging from the extreme left tail of the distribution (0.00005) to the extreme right tail (0.99995).

Once the set of quantile levels $\mathcal{T}$ is specified, the two-stage architecture proceeds to generate forecasts for each quantile through a sequence of rescaling operations. The second stage of the model generates quantile forecasts of the next month's raw stock returns $Q_{r_{i,t}}(\tau)$ for $\tau \in \mathcal{T}$. This stage consists of three steps. First, the standardised quantile forecasts $Q^{\mathcal{T}}_{\tilde{r}_{i,t}}$ from the first stage and the cross-sectional average of stock-level volatilities $\overline{\sigma}_t$ are used as inputs to rescale the standardised quantile forecasts back to the raw stock return scale $Q_{r_{i,t}}(\tau) = Q_{\tilde{r}_{i,t}}(\tau) \cdot \overline{\sigma}_t$. Second, the market-wide volatility sub-network generates a second multiplicative scaling factor to reintroduce a forward-looking aspect to the market-wide volatility scalar $\overline{\sigma}_t$. This market-wide volatility scaling factor is the same for all stocks in a given period t and is used to refine the quantile forecasts rescaled in the first step. Finally, both the standardised and raw quantile forecasts are clipped at -100% during the forward pass of the neural network.

In addition to the use of two stages, another key difference from the standard feed-forward neural network architecture is the inclusion of a bottleneck layer. Inspired by the autoencoder architecture, the bottleneck layer aims to capture the distribution hyperparameters. This is achieved by forcing the network to learn a compressed representation, as the bottleneck layer is limited to only four nodes.

The two-stage neural network takes three sets of inputs. The first set consists of 176 features representing stock-level characteristics based on 153 anomalies used by Tobek and Hronec (2021), 18 stock-level volatility estimates, and five cross-sectional averages of stock-level mean estimates. The second set of inputs consists of cross-sectional averages of the stock-level volatility characteristics used in the first set, i.e. 18 stock-level volatility estimates. The third input is a cross-sectional average of stock-level volatilities $\overline{\sigma}_t$ used for rescaling purposes. A detailed description of all features can be found in Section 3.1.1.

2.2. Training with the quantile loss function

In order to jointly predict multiple τ-quantiles using a two-stage neural network, individual τ-quantile losses must be aggregated into a single loss. Our model outputs both standardised and raw stock return quantiles, and we combine the respective losses using a simple average.

Let $\mathcal{B}$ denote a mini-batch consisting of B stock–month observations randomly sampled from the training panel data.[6] Each observation in $\mathcal{B}$ is indexed by a unique (i, t) pair, where i identifies the stock and t the month. The aggregated loss over the mini-batch is then defined as

$$\mathcal{L}^{\mathcal{T}} = \frac{1}{|\mathcal{B}|}\frac{1}{K}\sum_{\tau\in\mathcal{T}}\sum_{(i,t)\in\mathcal{B}}\left(\rho_\tau\left(r_{i,t+1} - \widehat{Q}_{r_{i,t+1}}(\tau|x_{i,t}, z_t, \overline{\sigma}_t)\right) + \rho_\tau\left(\tilde{r}_{i,t+1} - \widehat{Q}_{\tilde{r}_{i,t+1}}(\tau|x_{i,t}, z_t, \overline{\sigma}_t)\right)\right). \tag{3}$$

where $|\mathcal{B}| = B$ is the batch size, K is the number of quantiles in $\mathcal{T}$, $\widehat{Q}_{r_{i,t+1}}(\tau|x_{i,t}, z_t, \overline{\sigma}_t)$ is the model's predicted τ-quantile for the raw return of stock i at time $t+1$ conditional on firm characteristics $x_{i,t}$, market conditions z_t, and volatility scaling $\overline{\sigma}_t$, and $\widehat{Q}_{\tilde{r}_{i,t+1}}(\tau|x_{i,t}, z_t, \overline{\sigma}_t)$ is the predicted τ-quantile for the standardised return. The quantile loss function $\rho_\tau(\cdot)$ is defined as

$$\rho_\tau(\xi) = \mathbb{1}_{(\xi\geq 0)}\tau\xi + \mathbb{1}_{(\xi<0)}(\tau-1)\xi. \tag{4}$$

This piecewise linear function penalises underprediction and overprediction asymmetrically based on the quantile level τ, with equal weighting across stocks and quantiles in the aggregate loss.[7] Although $\rho_\tau(\cdot)$ is not differentiable at zero, modern autodiff frameworks such as PyTorch handle subgradients at non-differentiable points automatically.

It is important to note that each mini-batch $\mathcal{B}$ contains stock–month observations $(r_{i,t+1}, x_{i,t}, z_t, \overline{\sigma}_t)$ sampled from the entire training panel, where both the stock index i and the time index t vary across observations within the batch, with $x_{i,t}$ denoting the vector of firm-level characteristics, z_t the market-level variables, and $\overline{\sigma}_t$ the cross-sectional average volatility scaling factor. This means that a single batch spans multiple time periods and includes observations from many different stocks, allowing the model to aggregate information across both the cross-sectional and time-series dimensions of the panel. Because the model is trained on this panel with parameter sharing across stocks and time, it learns a conditional quantile function $Q_{r_{i,t+1}}(\tau|x_{i,t}, z_t, \overline{\sigma}_t)$ that depends on observable predictors and captures both cross-sectional heterogeneity (via stock characteristics $x_{i,t}$) and time variation (via market conditions z_t and the scaling factor $\overline{\sigma}_t$). Importantly, the quantile forecasts are time-varying even for a single stock, as both firm-level characteristics and market conditions evolve over time. Thus, $\widehat{Q}_{i,t}(\tau) \neq \widehat{Q}_{i,t'}(\tau)$ for $t \neq t'$ whenever the conditioning variables change.

To train the model, we minimise the loss function using stochastic gradient descent with mini-batches of size $B = 8192$ observations sampled from the training panel. Specifically, we use the Adam optimiser (Kingma & Ba, 2014) with an initial learning rate of 0.0003 and $(\beta_1, \beta_2) = (0.9, 0.999)$. We adjust the number of epochs based on the number of observations available at each training period. The number of epochs is set to 100 $\times$

[6] In stochastic gradient descent, a mini-batch is a subset of the complete training dataset that is processed together in a single forward and backward pass through the neural network. The full training dataset is divided into multiple mini-batches, and the model parameters are updated after processing each mini-batch.

[7] The linear combination of losses from the two output heads uses equal weights, though in practice, the standardised return component dominates, due to differences in scale. This design choice aligns with our modelling rationale, where the first stage captures most of the learning and the second stage refines the result.

(A/n), where n is the number of observations available for training and A is the adjustment constant equal to 3,000,000 for the full sample and 1,500,000 for the liquid sample, derived from the average number of observations in each sample over time.[8] We use early stopping with a patience of two epochs, where 20% of the training data are used as validation to determine the stopping point. After early stopping, the model state with the lowest validation loss is taken as the final model. We use Leaky ReLU (Maas et al., 2013) as the activation function in all layers except the last layer in each sub-network. We use batch normalisation (Ioffe & Szegedy, 2015) in all layers except the last layer in both the standardised τ-quantiles sub-network and the market-wide volatility sub-network. We apply dropout to all layers except the last layer in each sub-network and use a rate of 0.2 as a result of the hyperparameter search described in Appendix D. We also apply an L1 penalty to the weights of the first stage in each subnet. The L1 penalty for the first layer is 0.0001, and for the second layer, it is 0.00001. The first layer of the market-wide volatility network is also regularised by an L2 penalty of 0.00001. As an additional form of regularisation, we use an ensemble of 20 networks.

The model is trained from scratch for the first training period and then fine-tuned for subsequent training periods, meaning that weights from the previous training period are used to initialise the network instead of reinitialising randomly. Fine-tuning greatly reduces the computational burden of having to re-estimate the model each period and is a natural choice given the large overlap of subsequent training samples with the widening estimation window approach. It is expected that one year of new data should not change the estimated parameters too much unless there is a novel high-information event such as the dot-com bubble or the global financial crisis, in which case fine-tuning requires more epochs to retrain.

2.3. Approximating probability density and its moments from quantiles using B-splines

To approximate the probability density, let us consider a grid of τ-quantile forecasts $\{Q_j(\tau), \tau_j\}_{j=1}^{K}$, where τ_j is the probability and $Q_j(\tau)$ is the corresponding quantile value, containing K grid points that can be used to obtain the cumulative distribution function $F : \mathbb{R} \to [0, 1]$. Since stock returns range from -1 to ∞, from a practical perspective we are interested in $F : [-1, \infty) \to [0, 1]$. To get a denser representation, we use cubic B-splines[9] to interpolate between adjacent quantiles by constructing piecewise polynomial functions within each interval $[Q_j, Q_{j+1}]$. Instead of interpolating the predicted quantiles directly, we create a new denser $\mathcal{T}^d$ grid with 100 equally spaced points between every two original quantile prediction points, except the largest and smallest — i.e. the denser grid has $(K - 3) \times 100$ values — and fit the cubic B-spline to approximate the cumulative distribution function. To derive the probability density function, the derivative of the spline for the base function is calculated at each interval to obtain the new spline function representing the probability density function.

Fig. 2 shows an example of the interpolated cumulative distribution function on the left, and the implied probability density function on the right. The figure is based on τ-quantile forecasts of 22-day (monthly) returns for Microsoft in October 2008. The density function is relatively close to Student's t-distribution with noticeably large tails during the depths of the Great Financial Crisis. Fig. 3 then complements this with the evolution of the forecasted distribution of the 22-day-ahead returns over the period from 1996–2020.

As the interpolated density function does not cover the extreme tails of the distribution beyond the smallest and largest predicted quantile, these tails are modelled by discrete probabilities for values at the ends of the new $\mathcal{T}^d$ grid. We use discrete probabilities implied by the estimated spline function. The density function is truncated for returns less than -1, and the discrete probability is then considered for -1 returns. Excluding the extreme tails from the density function results in a slight underestimation of the variance and a more pronounced underestimation of the kurtosis of the distribution. Inferring the distribution in the tails from empirical data is problematic, so we choose this solution to avoid the problem to some extent. In cases where B-spline interpolation does not produce plausible results, we use a linear B-spline function as a fallback (see Appendix E.1 for details). A complete algorithm for approximating the probability density function from the quantile forecasts is described in detail in Appendix E, Algorithm 1.

Importantly, the B-spline approximation allows us to obtain moments of the cumulative probability distribution function. We first derive the z-th non-central moment of a density function $f(x)$:

$$m_z = \int_{-1}^{\infty} x^z f(x)\, dx, \tag{5}$$

using numerical integration methods and the B-spline-based density function approximation obtained above. We use a simple tractable local linear approximation algorithm, which leads to trivial integrals of polynomial functions. The complete algorithm is detailed in Appendix E, Algorithm 2. Further, using this approximation, we calculate central moments and rescale the moments by the integrated density function to ensure that they are properly defined.

The estimated empirical quantiles from neural networks do not cover the full range of possible stock returns because it is not possible to estimate quantiles in the extreme tails of the distribution. The calculated central moments are, therefore, biased, with the bias being greater for higher moments, as the tails of the distribution have a greater impact. It is, therefore, important to test how well the algorithm captures the true moments. To do this, we compare the true moments of a theoretical distribution with the output of the algorithm and propose an adjustment to reduce the measured bias. Appendix Section E.3 describes how to adjust the moments to account for bias.

[8] See Section 3.1 for the description of the data used.

[9] We use the splrep and BSpline functions from the Python SciPy library, Virtanen et al. (2020).

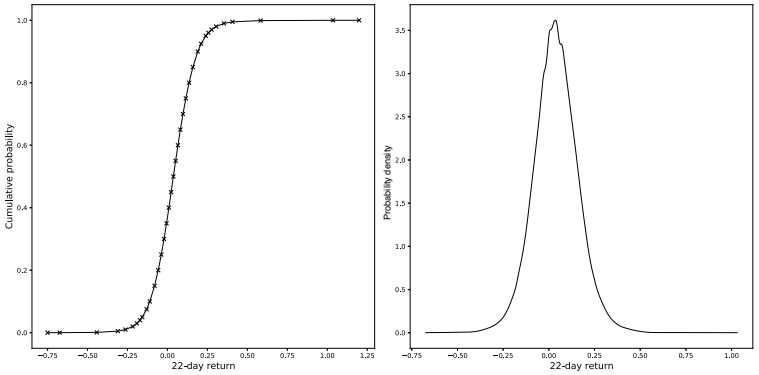


Fig. 2. Interpolated cumulative distribution function and probability density function. This figure shows the cumulative distribution function interpolated from the inverted quantile forecasts depicted as marks (left) and the derived probability density function (right) for out-of-sample 22-day-ahead returns of Microsoft stock (CUSIP 59491810) in October 2008. Quantile forecasts used to obtain the cumulative distribution function are generated using the two-stage quantile neural network model.

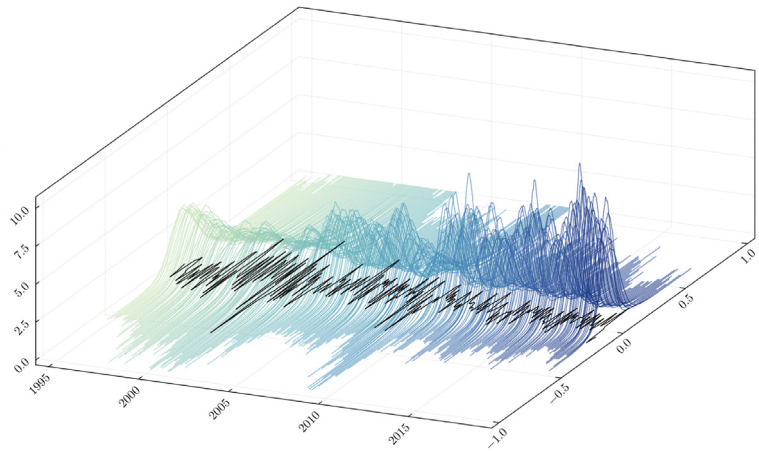


Fig. 3. Example of forecasted probability density function over time. This figure displays the forecasted probability density function for out-of-sample 22-day-ahead returns of Microsoft stock (CUSIP 59491810) generated using our two-stage quantile neural network model.

3. Empirical results

3.1. Data and features

Our data cover U.S. and international equity markets for the periods 1963 to 2018 and 1990 to 2018, respectively. We use the merged CRSP/Compustat database from the Wharton Research Data Service (WRDS) for the U.S. sample and LSEG Datastream[10] for the international sample. Data are pre-processed using the methodology described by Tobek and Hronec (2021). There are four regions in the sample — the U.S., Europe, Japan, and Asia-Pacific — which also includes 23 developed countries.[11] We report our results for both the *full sample* and the *liquid sample*, which includes only the most liquid stocks. The liquid sample includes stocks that meet the following criteria: a price greater than 1 (or 0.1 for Asia-Pacific) at the end of the previous month, trading volume within the top 95%, and trading volume within the top 95% of the total dollar trading volume over the previous 12 months in each region. The full sample includes stocks with a price greater than 1 ($0.1 for Asia-Pacific) at the end of the previous month. Market capitalisation and volume filters are not applied to the full sample. Micro-cap stocks in the full sample therefore account for only a small fraction of the capitalisation and trading volume of the total market. Table 1 summarises the minimum, average, and maximum number of stocks per month observed in the cross-section of each region over the sample period, separately for the full and liquid samples.

3.1.1. Features

The two-stage quantile neural network model uses both stock characteristics and market variables aggregated from the stock-level variables as features. There are 176 stock characteristics consisting of 153 anomalies from Tobek and Hronec (2021) plus an additional

[10] Formerly called Reuters or Refinitiv Datastream.

[11] The U.S., Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, the U.K., Japan, Australia, New Zealand, Hong Kong, and Singapore.

Table 1
Number of stocks in the cross-section across the regions. This table shows the average, minimum, and maximum number of stocks in the cross-section in each region for the full and liquid samples. The period covered is from 1973 to 2018 for the U.S. sample and from 1990 to 2018 for the international sample.

	Full sample				Liquid sample			
Region	USA	Europe	Japan	Asia-Pacific	USA	Europe	Japan	Asia-Pacific
min	3552	5098	1919	1157	873	413	534	226
mean	5136	5746	3225	2834	1167	690	749	430
max	7384	6740	3795	4553	1734	1044	1079	712

18 stock-level volatility estimates. The 23 market-level variables consist of 18 market volatility variables and five market mean variables. This brings the total number of features to 194.

The list of 153 anomalies is given in Appendix A and includes 93 fundamental, 49 market friction, and 11 I/B/E/S anomalies, mostly covered by McLean and Pontiff (2016), Hou et al. (2020), and Harvey et al. (2016). Compared to the original data, we use weekly data during training. This means that the fundamental signals are updated every week with financial statement information from financial years ending at least six months earlier. Trade data information, such as market capitalisation, is taken as the most recent available. Missing characteristic values are imputed using cross-sectional medians (Gu et al., 2020).

Each sub-network in the two-stage quantile neural network model uses its own set of features. The standardised τ-quantile sub-network uses 176 features consisting of 153 anomaly-based stock characteristics together with 18 stock-level volatilities and five market mean returns. The 18 stock-level volatilities are calculated using daily exponentially weighted moving averages (EWMAs) with $\alpha \in \{0.8, 0.9, 0.94, 0.96, 0.98, 0.99\}$ smoothing factors, EWMAs of negative returns with α smoothing factors of $\{0.8, 0.9, 0.94\}$ to proxy asymmetric volatility, total volatilities calculated over the past three, six, and 12 months, and Parkinson (1980) high–low range-based volatility. All stock-level volatility estimates are scaled by their cross-sectional mean to remove the effects of changes in market-wide average volatility. This is done to make stock-level volatilities more consistent with the other stock-level variables, which are normalised to cross-sectional empirical quantiles. The five additional market variables used as inputs to the standardised τ-quantile sub-network are computed as EWMAs of cross-sectional equally weighted averages of individual stock returns $\mu_t = \alpha \cdot \left(1/N_t \sum_{i=1}^{N_t} r_{i,t-1}\right) + (1-\alpha) \cdot \mu_{t-1}$ over the number of stocks N_t with smoothing parameters $\alpha \in \{0.9, 0.94, 0.96, 0.99, 0.999\}$. These variables measure the average historical market-wide return and should help in forecasting when there is some persistent time variance in expected market mean returns. We rescale the mean variables by the same scalar as the predicted returns to reflect the fact that their scale over time is relevant in the estimation.

The market-wide volatility sub-network uses 18 cross-sectional averages of stock-level volatilities described above. These serve as a proxy for market-wide index volatility, as they are likely to be relevant for estimating the distribution of individual stock returns, which includes both systematic and idiosyncratic volatility components.

3.1.2. Validation

We use the initial period from 1973 to 1994 for hyperparameter optimisation. Models with varying hyperparameters are trained from 1973 to 1989 and validated from 1990 to 1994. The hyperparameters yielding the best validation performance, measured in terms of average quantile loss (see Section 3.3 for details), are then applied to train models from 1995 to 2018, generating out-of-sample predictions. Detailed hyperparameter searches for both the full and liquid samples are documented in Appendix D together with the validation scheme.

From 1995 to 2018, using only the U.S. sample, we retrain the model annually using an expanding data window starting in 1973. To ensure that the 22-day-ahead return,[12] ends within the current calendar year, we restrict the final forecast date each year to November 29. All features are constructed using information available up to that point. This avoids using any information from the next year, thereby preventing both label and feature leakage from future data. Separate models are trained for full and liquid samples, with monthly out-of-sample predictions for both U.S. and international samples, using the most recently trained model at each month's end. Predictions from the full sample model are applied to the full sample and, similarly, to the liquid sample. Furthermore, to provide additional evidence of out-of-sample predictability robustness, we report predictability and profitability results for the period from 2019 to 2023 for the U.S. stocks in Appendix F.

3.2. Description of the estimated moments

Since our main empirical results are based on the predicted quantiles and moments of the distribution of all available stocks, we first describe the estimated moments for the U.S. data. As the rest of the international regions show similar dynamics, we present these results in Appendix Table C1. Specifically, we predict 37 τ-quantiles with $\tau \in \mathcal{T}$, where $\mathcal{T}$ is the predefined set $\mathcal{T} = \{0.00005, 0.0001, 0.001, 0.005, 0.01, \ldots, 0.05, 0.075, 0.1, 0.15, 0.2, \ldots, 0.8, 0.85, 0.9, 0.925, 0.95, \ldots, 0.99, 0.995, 0.999, 0.9999, 0.99995\}$, which provides denser coverage in the tails and sparser spacing in the centre of the distribution. This choice reflects a practical trade-off between statistical resolution and computational feasibility.[13] The

[12] Our forecast target.

[13] While using more quantiles is possible, it substantially increases computational and storage costs. We also experimented with fewer quantiles, but this led to inferior interpolation performance, particularly in the tails. Nonetheless, the architecture of the two-stage quantile neural network remains flexible and can accommodate changes to both the number and placement of quantiles—allowing, for example, denser tail coverage in applications such as value at risk.

Table 2
Summary statistics for the moments in the U.S. This table shows summary statistics for the forecasted first four moments for the full and liquid samples in the U.S. The moments are forecasted using the quantiles-to-moments algorithm introduced in Section 2.3. The table reports the standard deviation, skewness, and kurtosis without bias adjustment, described in Section E.3, as well as with the adjustment (Adj).

	Full sample			Liquid sample		
	Mean	Median	Std. Dev.	Mean	Median	Std. Dev.
Mean	0.0068	0.0089	0.0216	0.0098	0.0092	0.0126
Std. Dev.	0.1477	0.1317	0.0743	0.1071	0.0940	0.0539
Std. Dev. Adj	0.1487	0.1325	0.0750	0.1074	0.0944	0.0541
Skewness	0.8453	0.7572	0.6352	0.1800	0.1722	0.1993
Skewness Adj	0.9340	0.8205	0.7072	0.2068	0.1971	0.1839
Kurtosis	8.9784	7.5963	5.2384	5.4261	5.2248	2.3752
Kurtosis Adj	13.798	9.9362	19.8491	6.8452	6.2727	8.3637

grid is deliberately designed to provide denser coverage in the tails of the distribution, where spline interpolation is more fragile and tail behaviour is critical for applications such as risk management, while the central region is covered more sparsely (every 5%). This design also helps counteract the imbalance in loss aggregation. Since quantile losses are typically larger in the centre, uniform averaging across τ implicitly downweights the tails. Denser tail coverage helps restore balance and ensures that tail performance is better accounted for.

Further, we predict the mean, standard deviation, skewness, and kurtosis of all available individual stock returns. This pattern has practical relevance for portfolio construction. In the full sample, the high correlation implies that both characteristics yield similar asset ranking and thus similar products. In contrast, in the liquid sample, correlations close to zero support a potentially more diversified portfolio construction based on higher-order moments.

Table 2 contains summary statistics for the predicted moments, and Fig. 4 shows the histogram of the predicted mean, standard deviation, skewness, and kurtosis for the full and liquid samples for all stocks across the U.S.

We can see that the distribution of the mean for both samples is approximately centred around zero, with the liquid sample having a slightly higher density around the mean. The mean return is positive for both samples, with the liquid sample having a slightly higher mean (0.0098) than the full sample (0.0068). The standard deviation is higher for the full sample (0.1477) than for the liquid sample (0.1071) and shows a wider spread for the full sample than for the liquid sample. Skewness is significantly higher in the full sample (0.9340) than in the liquid sample (0.2068), reflecting more extreme positive and negative returns, while the liquid sample shows a more symmetrical distribution. Kurtosis is also higher in the full sample (13.798) than in the liquid sample (6.8452), indicating more pronounced tails compared to the liquid sample. Bias adjustment, described in Section E.3, has a large impact only on kurtosis, with negligible impact on the standard deviation, and a small impact on skewness.

Table 3 also includes average cross-sectional correlations between the first four predicted moments and the median of stock returns.[14] There is a high correlation of 0.95 between the forecasted mean and the median for the liquid sample, while the correlation is lower (0.75) for the full sample. This is consistent with the fact that the full sample contains more micro-cap stocks with higher skewness, which affects the mean but not the median. There is also a high correlation between skewness and kurtosis (0.91) in the full sample. In contrast, in the liquid sample, the correlation between skewness and kurtosis is close to zero. This pattern has practical relevance for portfolio construction. In the full sample, the high correlation implies that both characteristics yield similar asset ranking and thus similar products. In contrast, in the liquid sample, correlations close to zero support a potentially more diversified portfolio construction based on higher-order moments.

3.3. Predictability

This section evaluates the out-of-sample predictions of the quantiles from our two-stage quantile neural network. It also evaluates the predictions of the mean and variance estimated from the quantiles. This is of particular interest, as we argue that the first and second moments, estimated without any assumptions about the distribution, are more predictive of the mean and variance. To benchmark our forecasting strategy, we use a linear model (LNN), a single-hidden-layer neural network with 32 neurons (1hNN), a double-hidden-layer neural network with 128 neurons (2hNN), and the commonly used GARCH. Details of the benchmark models can be found in Appendix B.1 and Appendix B.2. We also compare the performance of the two-stage quantile model with tree-based machine learning benchmarks in Appendix I. In addition, we provide a simulation study in Appendix H analysing the performance of our forecasting strategy out of sample under a known data-generating process.

To measure predictive performance across the distribution, we report the average loss, which is defined as the time-series average of the cross-sectional τ-quantile losses, averaged across all τs for each month.

Specifically, we define the quantile loss for asset i at time t and quantile level τ as $L_{\tau,t,i} = \rho_\tau\left(r_{i,t+1} - \hat{Q}_\tau\right)$, as in Eq. (4), where $r_{i,t+1}$ denotes the realised return, and $\hat{Q}_\tau$ is the predicted τ-quantile from the model.

We then compute the overall average quantile loss as

$$L_{\text{avg}} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{1}{N_t}\sum_{i=1}^{N_t}\left(\frac{1}{K}\sum_{\tau\in\mathcal{T}} L_{\tau,t,i}\right)\right) \tag{6}$$

[14] The table reports results with bias adjustment described in Appendix Section E.3. The adjustment has only a small impact on the reported correlations.

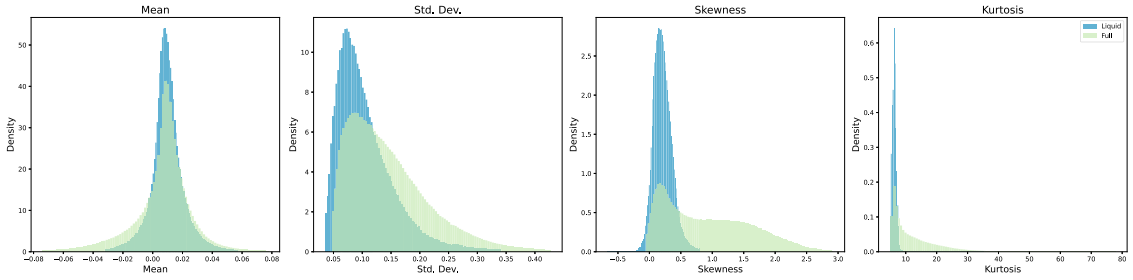


Fig. 4. Histogram of central distribution moments. This figure shows histograms of the forecasted first four moments for the full and liquid samples in the U.S. The moments are forecasted using the quantiles-to-moments algorithm introduced in Section 2.3, including the bias adjustment described in Section E.3. Displayed variables are trimmed to exclude the top and bottom 0.5% of their values.

Table 3
Average cross-sectional correlation between variables in the U.S. This table shows the average cross-sectional Spearman correlation between the forecasted first four moments and the median of the stock returns for the full and liquid samples in the U.S. Medians are forecasted using the two-stage model defined in Section 2, and moments are forecasted using the quantiles-to-density algorithm introduced in Section 2.3.

	Full sample					Liquid sample				
	Median	Mean	Variance	Skewness	Kurtosis	Median	Mean	Variance	Skewness	Kurtosis
Median	1.00	0.75	−0.63	−0.54	−0.43	1.00	0.95	−0.34	0.08	0.15
Mean	0.75	1.00	−0.22	−0.08	−0.08	0.95	1.00	−0.15	0.27	0.06
Var	−0.63	−0.22	1.00	0.52	0.22	−0.34	−0.15	1.00	0.65	−0.39
Skewness	−0.54	−0.08	0.52	1.00	0.91	0.08	0.27	0.65	1.00	−0.16
Kurtosis	−0.43	−0.08	0.22	0.91	1.00	0.15	0.06	−0.39	−0.16	1.00

Table 4
Out-of-sample quantile forecast evaluation. This table shows the average out-of-sample quantile cross-section loss for quantile forecasts from the two-stage model and alternative models. The results are obtained separately for the full and liquid samples for individual regions covering the out-of-sample period from 1995 to 2018. Neural network models are trained annually using the expanding window on U.S. data only, while out-of-sample forecasts are generated for both the U.S. and international samples. The average loss is calculated as the time-series average of the cross-sectional averages of the average quantile loss across all $\tau \in \mathcal{T}$, where $\mathcal{T}$ is defined in Section 3.2. The average loss is calculated as in Eq. (6) for the two-stage NN model, as well as for the benchmark neural network models: the linear model (LNN), the one-layer model (1hNN), the two-layer model (2hNN), and the benchmark GARCH model. Note that losses are multiplied by 100.0, and the t-statistics in parentheses are computed for the average loss difference between the given model specification and the two-stage benchmark model using Newey–West standard errors with 12 lags.

Specification	Variable	Full sample				Liquid sample			
		USA	Europe	Japan	Asia-Pacific	USA	Europe	Japan	Asia-Pacific
GARCH	L_{avg}	2.659	2.669	2.050	3.416	1.899	1.679	1.836	2.159
	(t-stat)	(8.47)	(14.07)	(8.15)	(15.51)	(6.38)	(4.49)	(3.05)	(6.23)
LNN	L_{avg}	2.592	2.520	2.020	3.143	1.874	1.658	1.820	2.115
	(t-stat)	(6.05)	(3.44)	(4.43)	(4.44)	(3.95)	(2.07)	(1.72)	(1.53)
1hNN	L_{avg}	2.569	2.496	2.004	3.120	1.869	1.655	1.819	2.121
	(t-stat)	3.81	(0.44)	(4.03)	(3.09)	(3.08)	(1.69)	(1.63)	(1.57)
2hNN	L_{avg}	2.564	2.494	2.001	3.123	1.870	1.654	1.823	2.128
	(t-stat)	(2.22)	(0.04)	(1.71)	(2.50)	(2.39)	(1.10)	(1.88)	(1.73)
Two-stage NN	L_{avg}	2.550	2.494	1.992	3.106	1.858	1.649	1.813	2.106

where T is the number of months, N_t is the number of stocks available at time t, $K = |\mathcal{T}|$ is the number of quantiles considered, and $L_{\tau,t,i}$ is the quantile loss defined above.

Table 4 shows the average out-of-sample τ-quantile losses for the full and liquid samples across all regions. The average losses of the two-stage quantile neural network are significantly smaller than all benchmark models when looking at the full sample data. Our two-stage model also significantly outperforms the 1hNN full sample model in all regions except Europe, and the 2hNN full sample model in the U.S. and Asia-Pacific. The average quantile loss is also lower in Japan, but the difference is not statistically significant at the 5% level.

Predictive performance differences for the liquid sample mirror those observed for the full sample, although to a lesser extent and with fewer statistically significant differences. As expected, the average out-of-sample quantile losses are generally lower for the liquid sample than for the full sample, reflecting the lower volatilities of the underlying stocks. The two-stage model consistently outperforms the GARCH model across all regions. However, when compared to alternative neural network specifications, the two-stage model only shows significant outperformance in the U.S. region. Internationally, while the two-stage model achieves lower average quantile losses than the alternative neural network specifications, these differences do not reach statistical significance at the 5% level, except for the 1hNN model in Europe. In addition

Table 5
Mean forecast evaluation. This table shows the R^2 of the mean and median forecasts of the next 22 days of stock returns for the full and liquid samples across all regions. The mean prediction based on the two-stage quantile model (Two-stage NN) is compared to the direct mean prediction from the two-hidden-layer neural network (2hNN MSE — Mean) and the median prediction from the two-stage model (Two-stage NN — Median). The results are obtained separately for full and liquid samples on individual regions covering the period from 1995 to 2018. Models are retrained annually using the expanding window for U.S. data only, while out-of-sample forecasts are generated for both U.S. and international data. Note that losses are multiplied by 100, and values in parentheses are Diebold–Mariano test statistics for the difference between the model in a row and the two-stage NN mean. Standard errors are adjusted using the Newey–West standard deviation estimator with 12 lags.

Specification	Full sample					Liquid sample				
	USA	Europe	Japan	Asia-Pacific	Global	USA	Europe	Japan	Asia-Pacific	Global
2hNN MSE — Mean	0.65	0.14	0.85	0.69	0.51	−0.68	−0.44	−0.48	−0.77	−0.54
	(−2.54)	(−3.56)	(−2.37)	(−3.39)	(−3.55)	(−1.81)	(−1.31)	(−1.84)	(−1.79)	(−1.98)
Two-stage NN — Median	0.10	−0.04	0.30	0.06	0.09	0.37	0.32	0.47	0.55	0.50
	(−3.50)	(−4.92)	(−2.82)	(−4.83)	(−4.93)	(0.23)	(0.27)	(0.98)	(0.39)	(0.52)
Two-stage NN — Mean	1.63	0.87	1.66	1.63	1.36	0.32	0.22	0.18	0.40	0.37

to aggregated quantile loss, we also report the τ-specific out-of-sample quantile losses in Table G1 and the continuous ranked probability score in Table G2 in Appendix G.

These performance differences also highlight the role of model flexibility in capturing complex distributional patterns. In particular, the consistent improvement from LNN to 2hNN across regions reflects the benefits of modelling non-linearities in the conditional quantile functions. The two-stage model, which incorporates additional structural layers and nonlinear interactions across features and volatilities, achieves further gains — especially in the full sample — suggesting that the cross-sectional distribution of stock returns depends on non-linear effects that are not well captured by linear architectures.

We also examine the performance of the mean and variance forecasts. To compare our two-stage model and the forecasts obtained by the algorithms described above, we compute the mean forecasts directly using a two-hidden-layer neural network model estimated with a mean square error loss function. We follow (Gu et al., 2020) and modify[15] the calculation of the Diebold–Mariano test statistic to compare the cross-sectional average of forecast errors instead of comparing individual errors.

Table 5 shows the out-of-sample R^2 of the mean and median forecasts. Our two-stage model provides the highest out-of-sample R^2. For the liquid sample and across all regions, the highest R^2 is consistently achieved by the median forecast from the two-stage model. Forecasts using the median are not statistically different from those using the mean in the two-stage model. The R^2 for direct mean predictions from the two-hidden-layer neural network is negative and smaller than the two-stage predictions. In the case of the full sample, the highest R^2 is consistently achieved by the mean prediction derived from our model.

The results reflect a trade-off between robustness to extremes and sensitivity to the full shape of the conditional distribution. Quantile loss, by design, is less sensitive to large forecast errors, and the resulting median forecast is robust to outliers and heavy tails. However, this robustness can come at the cost of discarding potentially informative variation in the tails, which are particularly influential when forecasting the conditional mean—a non-robust functional. In the full sample, where extreme returns are more prevalent, this information appears valuable, and the mean forecast derived via numerical integration over the estimated quantile distribution outperforms both the median and the MSE-based benchmark. Notably, even the direct MSE-trained mean from the 2hNN achieves a higher R^2 than the median in this setting, highlighting that robustness does not necessarily translate into superior predictive performance when the tails contain signals. In contrast, the liquid sample exhibits thinner tails and fewer extreme returns. In this setting, the benefits of robustness are more pronounced, and the median forecast achieves competitive or superior performance. Importantly, since both the mean and median are derived from the same underlying quantile forecasts in our two-stage model, the comparison isolates differences attributable to the target functional rather than the model class. These findings underscore that the relative performance of robust versus non-robust functionals depends on the tail behaviour of the return distribution and the presence (or absence) of influential observations.

Finally, to evaluate the second-moment forecasts, we use the mean absolute deviation (MAD) and the root mean squared error (RMSE) of the predicted 22-day volatility for the full and liquid samples for all regions. Realised volatility is calculated as the square root of the sum of squared daily returns over the 22 trading days. Table 6 shows a highly significant outperformance of the volatility forecast based on the two-stage model in terms of both the MAD and RMSE for all regions and for both the full and liquid samples.

To clarify the source of improvement, we emphasise that the accuracy of the empirical moment estimates — such as the mean and variance — primarily depends on the quality of the underlying distributional forecasts. While we rely on numerical integration over interpolated quantile forecasts, this approach is not tied to our two-stage neural network architecture. In principle, any sufficiently accurate quantile forecasting method, whether traditional or machine learning-based, can be used in conjunction with this integration technique to obtain distributional moments. However, our empirical results show that the two-stage QNN consistently delivers more accurate quantile forecasts than alternative models, particularly in the full sample. Overall, we demonstrated the superior distributional forecasting performance of the

[15] See Appendix G for the description of the testing procedure.

Table 6
Volatility forecast evaluation. This table shows the mean absolute deviation (MAD), root mean squared error (RMSE), and Diebold–Mariano test statistic (DM) comparing the corresponding losses of the two competing models for the 22-day volatility forecasts. Volatility forecasting based on two-stage quantile forecasting (two-stage NN) is compared with GARCH model forecasting. The results are obtained separately for full and liquid samples on individual regions covering the period from 1995 to 2018. Neural network models are retrained annually using an expanding window on U.S. data only, while out-of-sample forecasts are generated for both U.S. and international samples. Diebold–Mariano test statistics are computed for the difference between the GARCH and two-stage NN model volatility forecasts. Note that losses are multiplied by 100, and the standard errors used in the Diebold–Mariano test statistics are adjusted using the Newey–West standard deviation estimator with 12 lags.

Specification	Variable	Full sample					Liquid sample				
		USA	Europe	Japan	Asia-Pacific	Global	USA	Europe	Japan	Asia-Pacific	Global
GARCH	MAD	7.34	8.79	7.79	11.1	8.53	3.89	3.54	3.80	4.87	3.93
Two-stage NN	MAD	4.63	5.20	3.99	6.49	4.99	2.93	2.63	2.81	3.16	2.87
GARCH	RMSE	13.14	16.24	14.63	18.70	15.50	6.93	7.10	6.72	9.52	7.36
Two-stage NN	RMSE	7.46	8.52	5.64	10.05	8.08	4.62	4.03	3.98	4.86	4.38
	DM	(10.25)	(35.92)	(20.32)	(44.26)	(26.47)	(7.90)	(5.98)	(7.97)	(9.40)	(10.02)

two-stage model and the moments derived from it compared to alternative models. The average out-of-sample quantile loss represents the forecast over the entire distribution, while the evaluation of the mean and variance forecasts provides additional evidence derived from the forecasted quantiles. While our evaluation focuses on distributional forecast accuracy and central moments, the underlying quantile estimates are also applicable to risk management tasks such as value-at-risk (VaR) forecasting. A formal backtesting exercise in this context is beyond the scope of this paper, but we acknowledge its practical relevance and view it as a promising direction for future work.

3.4. Distribution and the cross-section of stock returns

In addition to forecasting, we aim to investigate whether individual distributional characteristics are priced into the cross-section of returns through the profitability of long-short portfolios formed based on forecasted τ-quantiles as well as the profitability of long-short portfolios formed using central moments: mean, volatility, skewness, and kurtosis.

3.4.1. τ-Quantiles and the cross-section of stock returns

Table 7 shows the average monthly returns and annualised Sharpe ratios for long-short equally weighted decile portfolios constructed using the τ-quantile forecasts of the two-stage model. Each month, the long and short legs of the portfolios are constructed by selecting the top and bottom 10% of stocks with the highest and lowest forecasts, respectively.

The average monthly returns and Sharpe ratios are the highest for the long-short portfolios formed based on the central τ-quantiles, τs of 50% or 60%, for both the full sample and the liquid sample. Average monthly returns decrease as we move towards the tails of the distribution. For the full sample, average monthly returns are not significantly different from zero for τs greater than 80% and less than 10%. For the liquid sample, the average monthly returns fall even more sharply as one moves towards the tails of the distribution, becoming not significantly different from zero for τs greater than 70% and less than 40%. The highest Sharpe ratios are achieved by the long-short portfolios with the highest average monthly returns.

These results suggest that the tail τ-quantiles in isolation are not priced into the cross-section of stock returns. On the contrary, the central τ-quantiles are associated with the highest average returns. We also provide international evidence showing similar results in Table C2. To further investigate the role of the distribution in pricing the cross-section of stock returns, we turn to the moments of the distribution in the next section.

3.4.2. Distributional moments and the cross-section of stock returns

We then examine the profitability of long-short portfolios formed on the basis of the central moments of the distribution — mean, variance, skewness, and kurtosis — forecasts estimated from our two-stage quantile model. To illustrate the importance of asymmetry in the data, we first compare the profitability of portfolios formed on the mean and median forecasts. Table 8 shows the average monthly returns and Sharpe ratios of the long-short decile portfolios based on the mean and median forecasts across all regions.

To benchmark our forecasts, we use the mean forecasts obtained directly from the feed-forward neural network model estimated with the mean squared error loss function, as well as its distributional forecast equivalent, where the mean forecast is derived from predicted quantiles. The long-short portfolio based on the mean forecast of our two-stage quantile model leads to significantly higher average monthly returns and Sharpe ratios compared to the portfolios based on the mean forecasts obtained directly from the simple neural network for the full sample across all regions. This is consistent with the R^2 results in Table 5, as our two-stage model outperforms the direct mean forecast in terms of the R^2 across all regions and also leads to higher corresponding profitability. In the case of the full sample, the average monthly returns of our mean forecast are economically higher than the direct mean forecast in all regions and significantly higher for Japan, Asia-Pacific, and globally.

Next, the average monthly returns of the mean- and median-based long-short portfolios are statistically indistinguishable for the liquid sample. This is also consistent with the R^2 results in Table 5, as there are no significant differences in the quality of the mean and median

Table 7
Quantiles and the cross-section of stock returns in the U.S. This table shows the average and volatility of monthly returns and annualised Sharpe ratios for long, short, and long-short equally weighted decile portfolios over the period from 1995 to 2018. For each τ-quantile forecast, the long portfolio is formed by taking the top 10% of stocks with the highest forecast each month, The short portfolio is formed by taking the bottom 10% of stocks with the lowest forecast each month. The long-short portfolio is created by taking the difference between the long and short portfolios. Values in brackets are t-statistics adjusted for Newey–West standard errors with 12 lags. Results are obtained separately for the full and liquid samples for the U.S. only.

	Full sample								
	Long			Short			Long-short		
τ	Mean	Vol	SR	Mean	Vol	SR	Mean	Vol	SR
0.01	1.22 (5.62)	2.53	1.67	0.14 (0.19)	12.12	0.04	1.09 (1.52)	12.18	0.31
0.05	1.25 (5.66)	2.46	1.76	−0.09 (−0.12)	15.59	−0.02	1.34 (1.86)	11.90	0.39
0.1	1.28 (5.73)	2.48	1.79	−0.26 (−0.35)	12.87	−0.07	1.53 (2.13)	11.78	0.45
0.2	1.34 (6.04)	2.48	1.87	−0.57 (−0.78)	12.34	−0.16	1.91 (2.68)	11.61	0.57
0.3	1.46 (6.54)	2.58	1.96	−0.91 (−1.27)	11.68	−0.27	2.37 (3.36)	11.09	0.74
0.4	1.88 (7.44)	3.16	2.06	−1.34 (−1.91)	11.05	−0.42	3.22 (4.89)	9.96	1.12
0.5	2.88 (6.97)	5.60	1.78	−1.82 (−2.68)	10.17	−0.62	4.70 (8.25)	7.17	2.27
0.6	3.44 (5.68)	8.70	1.37	−2.12 (−3.51)	8.35	−0.88	5.56 (9.41)	4.86	3.96
0.7	3.48 (4.73)	11.16	1.08	−1.42 (−3.57)	4.47	−1.10	4.90 (6.57)	8.53	1.99
0.8	2.98 (3.72)	12.59	0.82	0.44 (1.65)	2.46	0.62	2.55 (3.16)	11.78	0.75
0.9	2.12 (2.57)	13.35	0.55	0.75 (3.29)	2.36	1.10	1.37 (1.69)	12.49	0.38
0.95	1.71 (2.11)	13.16	0.45	0.86 (4.04)	2.44	1.22	0.85 (1.07)	12.80	0.23
0.99	1.36 (1.70)	13.09	0.36	0.97 (5.04)	2.75	1.22	0.39 (0.50)	12.28	0.11
	Liquid sample								
	Long			Short			Long-short		
τ	Mean	Vol	SR	Mean	Vol	SR	Mean	Vol	SR
0.01	0.95 (5.26)	2.99	1.10	0.16 (0.21)	11.09	0.05	0.79 (1.06)	11.40	0.24
0.05	0.96 (5.63)	2.89	1.15	0.15 (0.20)	12.99	0.04	0.81 (1.11)	11.22	0.25
0.1	0.98 (5.72)	2.90	1.17	0.11 (0.14)	12.70	0.03	0.87 (1.20)	11.59	0.26
0.2	1.02 (5.89)	2.92	1.21	0.04 (0.06)	13.86	0.01	0.98 (1.34)	11.32	0.30
0.3	1.10 (6.09)	3.05	1.25	−0.04 (−0.06)	13.86	−0.01	1.14 (1.58)	10.97	0.36
0.4	1.39 (6.31)	3.59	1.34	−0.14 (−0.20)	12.12	−0.04	1.53 (2.23)	9.81	0.54
0.5	1.86 (5.61)	5.86	1.10	−0.35 (−0.53)	10.10	−0.12	2.21 (4.21)	6.33	1.21
0.6	1.74 (3.69)	8.49	0.71	−0.38 (−0.88)	6.27	−0.21	2.11 (5.40)	5.45	1.34
0.7	1.66 (2.86)	10.27	0.56	0.48 (2.50)	3.08	0.54	1.18 (2.23)	9.51	0.43
0.8	1.28 (1.94)	11.37	0.39	0.71 (3.95)	2.89	0.85	0.57 (0.91)	10.97	0.18
0.9	0.91 (1.24)	11.68	0.27	0.76 (4.29)	2.83	0.93	0.15 (0.21)	10.39	0.05
0.95	0.77 (1.03)	11.60	0.23	0.80 (4.62)	2.83	0.98	−0.03 (−0.04)	10.39	−0.01
0.99	0.60 (0.80)	12.23	0.17	0.84 (4.72)	2.88	1.01	−0.23 (−0.32)	11.38	−0.07

Table 8
Mean and median portfolios. This table shows average monthly returns and annualised Sharpe ratios for long-short equally weighted decile portfolios based on the median and mean predictions from the two-stage quantile NN model, the mean prediction from the two-hidden-layer neural network with multi-head quantile loss (2hNN Quantile − Mean), and the mean prediction from the two-hidden-layer neural network with MSE loss (2hNN MSE − Mean). The results are shown for the full and liquid samples and for individual regions, covering the period from 1995 to 2018. The long portfolio is formed by taking the top 10% of stocks with the highest forecast each month, and the short portfolio is formed by taking the bottom 10% of stocks with the lowest forecast each month. Values in parentheses are t-statistics adjusted for Newey–West standard errors with 12 lags. The significance of the difference in mean returns is tested using t-statistics with Newey–West standard errors with 12 lags.

Specification	Variable	Full sample					Liquid sample				
		USA	Europe	Japan	Asia-Pacific	Global	USA	Europe	Japan	Asia-Pacific	Global
2hNN MSE − Mean	Avg Ret	5.00	3.28	3.11	5.71	4.27	1.62	1.71	1.67	2.05	1.76
	SR	3.47	2.87	2.52	3.72	5.50	0.87	1.63	1.52	1.38	2.01
	t-stat	4.78	9.26	5.78	5.79	9.93	1.87	1.53	2.81	2.30	2.54
2hNN Quantile − Mean	Avg Ret	5.59	4.17	3.44	6.43	4.91	1.67	1.86	1.68	2.22	1.86
	SR	4.10	3.70	2.86	4.28	6.35	0.86	1.80	1.64	1.51	2.09
	t-stat	3.70	6.01	4.20	6.26	7.92	1.71	0.68	2.09	1.90	1.93
Two-stage NN − Median	Avg Ret	4.70	3.35	3.10	5.07	4.05	2.21	2.00	2.25	2.95	2.35
	SR	2.27	2.48	1.84	3.06	3.53	1.21	1.52	1.51	1.82	2.03
	t-stat	4.26	7.14	5.62	8.75	9.16	0.84	0.19	0.10	1.07	0.75
Two-stage NN − Mean	Avg Ret	6.44	5.29	4.03	8.05	5.95	2.35	2.02	2.27	3.12	2.44
	SR	4.34	4.49	3.31	4.99	6.92	1.70	1.88	1.92	2.13	2.83

Table 9
Moments decile portfolios. This table shows average monthly returns for individual deciles (1–10) and long-short equally weighted portfolios built at the predicted moments. Results are reported for the full and liquid samples, as well as for individual regions, covering the period from 1995 to 2018. Values in parentheses are t-statistics adjusted for Newey–West standard errors with 12 lags.

Mean	Full sample				Liquid sample			
Decile	USA	Europe	Japan	Asia-Pacific	USA	Europe	Japan	Asia-Pacific
1	−2.37 (−3.87)	−1.06 (−1.87)	−1.24 (−1.99)	−1.69 (−2.40)	−0.48 (−0.81)	−0.52 (−0.90)	−1.13 (−1.99)	−1.23 (−1.73)
2	−0.46 (−0.97)	−0.29 (−0.63)	−0.40 (−0.77)	−0.31 (−0.53)	0.33 (0.80)	0.18 (0.44)	−0.32 (−0.65)	−0.00 (−0.00)
3	0.33 (0.90)	0.04 (0.10)	−0.19 (−0.41)	0.13 (0.23)	0.64 (2.00)	0.51 (1.32)	−0.01 (−0.02)	0.48 (0.97)
4	0.68 (2.07)	0.35 (0.93)	0.10 (0.24)	0.46 (0.87)	0.75 (2.56)	0.71 (1.97)	0.16 (0.38)	0.58 (1.25)
5	1.01 (3.39)	0.57 (1.64)	0.38 (0.94)	0.87 (1.69)	0.90 (3.21)	0.72 (2.03)	0.20 (0.50)	0.68 (1.53)
6	1.28 (4.37)	0.92 (2.67)	0.70 (1.82)	1.30 (2.57)	1.00 (3.77)	0.91 (2.70)	0.34 (0.96)	1.04 (2.59)
7	1.46 (4.94)	1.22 (3.38)	0.88 (2.27)	1.68 (3.27)	1.17 (4.17)	0.94 (2.57)	0.47 (1.27)	1.10 (2.61)
8	1.82 (5.50)	1.51 (4.05)	1.27 (3.05)	2.30 (4.30)	1.20 (3.77)	1.13 (3.06)	0.65 (1.74)	1.26 (2.83)
9	2.30 (5.92)	2.05 (5.18)	1.63 (3.68)	3.05 (5.02)	1.44 (4.49)	1.37 (3.50)	0.91 (2.45)	1.52 (3.18)
10	4.07 (6.72)	4.23 (8.55)	2.79 (5.44)	6.36 (8.66)	1.87 (5.03)	1.50 (3.55)	1.14 (2.89)	1.88 (3.68)
10–1	6.44 (9.76)	5.29 (12.36)	4.03 (10.26)	8.05 (18.56)	2.35 (5.16)	2.02 (5.51)	2.27 (6.90)	3.11 (7.89)
Median								
1	−1.82 (−2.68)	−0.42 (−0.62)	−0.96 (−1.40)	−0.99 (−1.21)	−0.35 (−0.54)	−0.51 (−0.80)	−1.10 (−1.76)	−1.09 (−1.43)
2	0.16 (0.29)	0.52 (1.00)	0.08 (0.13)	1.01 (1.42)	0.34 (0.73)	0.23 (0.51)	−0.33 (−0.63)	−0.10 (−0.18)
3	0.65 (1.48)	0.29 (0.66)	0.25 (0.53)	1.28 (2.01)	0.61 (1.75)	0.50 (1.28)	−0.03 (−0.07)	0.50 (1.01)
4	0.99 (2.68)	0.50 (1.31)	0.28 (0.64)	1.14 (1.99)	0.75 (2.44)	0.69 (1.84)	0.17 (0.40)	0.48 (1.00)
5	1.19 (3.74)	0.69 (1.99)	0.36 (0.92)	1.14 (2.23)	0.83 (2.93)	0.77 (2.19)	0.29 (0.78)	0.65 (1.46)
6	1.29 (4.45)	0.91 (2.82)	0.53 (1.44)	1.08 (2.24)	1.08 (3.91)	0.91 (2.67)	0.22 (0.66)	0.93 (2.20)
7	1.44 (5.51)	1.09 (3.38)	0.82 (2.18)	1.37 (2.95)	1.13 (4.29)	0.95 (2.76)	0.46 (1.25)	1.16 (2.81)
8	1.52 (5.26)	1.36 (3.92)	1.04 (2.72)	1.82 (3.83)	1.18 (4.07)	1.03 (2.83)	0.65 (1.87)	1.39 (3.31)
9	1.80 (6.00)	1.65 (4.58)	1.34 (3.54)	2.22 (4.40)	1.41 (4.81)	1.39 (3.80)	0.92 (2.61)	1.53 (3.53)
10	2.88 (6.97)	2.94 (6.73)	2.14 (4.62)	4.08 (6.56)	1.86 (5.61)	1.49 (3.73)	1.16 (3.17)	1.86 (3.81)
10–1	4.70 (8.25)	3.35 (6.64)	3.10 (6.76)	5.07 (10.76)	2.21 (4.22)	2.00 (4.65)	2.25 (5.88)	2.96 (6.27)
Volatility								
1	1.02 (4.96)	0.75 (3.23)	0.25 (0.98)	0.76 (2.50)	0.88 (5.13)	0.81 (3.54)	0.33 (1.57)	1.00 (3.14)
2	1.11 (4.28)	0.83 (2.87)	0.36 (1.18)	1.11 (2.89)	0.98 (4.19)	0.84 (2.88)	0.26 (0.90)	0.95 (2.80)
3	1.08 (3.97)	0.92 (2.85)	0.49 (1.44)	1.07 (2.40)	1.02 (4.02)	0.85 (2.84)	0.32 (0.95)	0.91 (2.22)
4	1.09 (3.82)	0.85 (2.38)	0.59 (1.55)	0.98 (1.93)	1.10 (4.09)	0.82 (2.50)	0.27 (0.80)	0.97 (2.20)
5	1.05 (3.24)	0.85 (2.30)	0.60 (1.44)	0.95 (1.73)	0.98 (3.49)	0.81 (2.16)	0.50 (1.38)	0.97 (2.16)
6	1.05 (2.89)	0.66 (1.65)	0.62 (1.39)	0.96 (1.70)	0.96 (3.25)	0.82 (2.11)	0.36 (0.95)	0.79 (1.52)
7	1.05 (2.44)	0.57 (1.25)	0.65 (1.31)	1.08 (1.72)	0.91 (2.41)	0.74 (1.79)	0.26 (0.56)	0.62 (1.24)
8	0.86 (1.65)	0.37 (0.73)	0.83 (1.48)	1.00 (1.37)	0.84 (1.89)	0.84 (1.82)	0.31 (0.64)	0.71 (1.24)
9	0.74 (1.16)	0.62 (1.04)	0.89 (1.41)	1.61 (1.94)	0.72 (1.29)	0.71 (1.33)	0.13 (0.22)	0.62 (0.90)
10	1.05 (1.33)	3.12 (4.21)	0.61 (0.85)	4.63 (4.99)	0.44 (0.58)	0.23 (0.32)	−0.34 (−0.45)	−0.24 (−0.31)
10–1	0.03 (0.04)	2.36 (3.35)	0.35 (0.62)	3.86 (5.10)	−0.44 (−0.60)	−0.57 (−0.96)	−0.68 (−1.05)	−1.24 (−1.81)
Skewness								
1	0.85 (3.00)	0.73 (2.07)	0.31 (0.96)	0.69 (1.53)	0.61 (2.65)	0.30 (0.90)	−0.06 (−0.19)	0.44 (1.19)
2	1.08 (4.01)	0.88 (2.27)	0.37 (1.04)	0.84 (1.76)	0.75 (2.82)	0.45 (1.20)	−0.04 (−0.11)	0.46 (1.07)
3	1.05 (3.50)	0.83 (1.99)	0.48 (1.12)	0.77 (1.50)	0.77 (2.74)	0.58 (1.66)	0.06 (0.14)	0.59 (1.38)
4	1.03 (3.13)	0.61 (1.45)	0.50 (1.08)	0.67 (1.20)	0.81 (2.93)	0.79 (2.26)	0.19 (0.48)	0.70 (1.55)
5	1.04 (3.04)	0.51 (1.21)	0.55 (1.13)	0.97 (1.70)	0.84 (2.94)	0.78 (2.18)	0.20 (0.48)	0.86 (1.85)
6	0.88 (2.51)	0.59 (1.50)	0.56 (1.12)	1.31 (2.26)	1.02 (3.23)	0.84 (2.47)	0.42 (1.03)	0.74 (1.52)
7	0.90 (2.27)	0.71 (1.74)	0.67 (1.38)	1.67 (2.70)	1.11 (3.45)	1.00 (2.79)	0.38 (0.85)	1.02 (2.15)
8	0.96 (2.18)	0.99 (2.32)	0.87 (1.74)	2.20 (3.02)	1.09 (2.95)	0.97 (2.35)	0.52 (1.20)	1.01 (2.05)
9	1.23 (2.45)	1.65 (3.83)	0.98 (1.92)	2.67 (3.65)	1.03 (2.34)	1.04 (2.26)	0.54 (1.25)	0.95 (1.54)
10	1.09 (1.92)	2.03 (4.92)	0.61 (1.30)	2.37 (3.63)	0.80 (1.10)	0.72 (1.16)	0.19 (0.35)	0.54 (0.75)
10–1	0.24 (0.57)	1.30 (4.07)	0.30 (1.08)	1.68 (4.06)	0.18 (0.28)	0.42 (1.02)	0.25 (0.72)	0.11 (0.20)
Kurtosis								
1	1.08 (2.69)	0.82 (1.90)	0.48 (1.01)	0.80 (1.46)	0.76 (1.26)	0.39 (0.61)	0.03 (0.04)	0.47 (0.70)
2	0.99 (3.32)	0.93 (2.28)	0.47 (1.26)	0.68 (1.34)	0.70 (1.44)	0.67 (1.34)	−0.17 (−0.32)	0.45 (0.70)
3	1.00 (3.35)	0.76 (1.76)	0.51 (1.25)	0.69 (1.28)	0.70 (1.85)	0.63 (1.44)	0.11 (0.24)	0.34 (0.61)
4	1.05 (3.40)	0.69 (1.62)	0.49 (1.11)	0.90 (1.58)	0.80 (2.43)	0.73 (1.98)	0.23 (0.57)	0.72 (1.57)
5	0.96 (2.76)	0.54 (1.29)	0.57 (1.19)	1.07 (1.93)	0.78 (2.49)	0.90 (2.48)	0.37 (0.99)	0.76 (1.68)
6	0.99 (2.64)	0.72 (1.68)	0.70 (1.37)	1.78 (2.83)	0.91 (3.19)	0.78 (2.27)	0.43 (1.32)	0.99 (2.34)
7	0.99 (2.27)	0.84 (2.05)	0.74 (1.51)	1.94 (2.91)	1.13 (4.45)	0.90 (2.71)	0.50 (1.46)	1.00 (2.30)
8	1.02 (2.30)	1.31 (3.09)	0.91 (1.82)	2.36 (3.28)	1.07 (3.98)	0.90 (2.67)	0.37 (1.10)	0.88 (2.12)
9	1.19 (2.42)	1.49 (4.03)	0.76 (1.62)	2.41 (3.68)	1.07 (3.77)	0.94 (2.85)	0.37 (1.07)	1.02 (2.29)
10	0.83 (2.16)	1.42 (4.54)	0.27 (0.64)	1.52 (2.95)	0.92 (3.37)	0.62 (1.90)	0.16 (0.47)	0.69 (1.63)
10–1	−0.25 (−0.77)	0.60 (1.82)	−0.21 (−0.66)	0.73 (1.80)	0.15 (0.33)	0.23 (0.62)	0.14 (0.33)	0.22 (0.57)

forecasts for the liquid sample. However, the results are different for the full sample. The difference between the mean and the median is associated with distributional asymmetry, which is more pronounced for stocks in the full sample. Mean-based long-short portfolios have significantly higher average monthly returns than median-based long-short portfolios. This is true for all regions and also globally. The difference between mean and median

pricing for the full sample indicates the effect of forecast distributional asymmetry in the cross-section of stock returns.

In addition to examining the role of individual τ-quantiles and the mean, we look at the decile portfolios and the corresponding long-short portfolios formed on the basis of the higher moments. The higher moments are obtained by the quantiles-to-moments algorithm using the two-stage model quantile forecasts, as described in Section 2.3. Table 9 shows the average monthly returns and Sharpe ratios of the equally weighted long-short decile portfolios based on the forecast mean, median, volatility, skewness, and kurtosis.

Overall, none of the higher moments appear to be monotonically priced in the cross-section of stock returns in the liquid or full sample. Average monthly returns of the long-short portfolios formed on the higher moments are mostly not significantly different from zero. There are some regional exceptions for the full sample, where volatility- and skewness-based long-short portfolios have significantly positive average monthly returns in Europe and Asia-Pacific. The results for the value-weighted long-short portfolios are mostly consistent with the equally weighted portfolios and are shown in Table C3 in Appendix.

The relationship between predicted volatility and future returns is negative in the liquid sample but statistically insignificant, reflecting the overall ambiguity documented in prior studies on the pricing of volatility risk (Ang, Hodrick, Xing, & Zhang, 2006; Bollerslev et al., 2020). This result is consistent with the focus on large-cap stocks, where volatility effects are typically weaker. In the full sample, microstructure-related distortions in equally weighted portfolios further obscure the relationship, although the pattern becomes somewhat clearer with value weighting.

The results for skewness are inconclusive for a monotonic relationship with future returns. However, there is a noticeable hill-shaped effect: stocks with low skewness have lower future returns relative to stocks with more average skewness. The low returns of the high-skewness portfolio (10) lead to a non-monotonic relationship across the decile portfolios and thus also affect the 10-1 long-short portfolio. Skewness would be more convincingly priced if 9-1 long-short portfolios were used instead.

4. Conclusion

We developed a novel machine learning approach to forecasting the distribution of stock returns, focusing on both U.S. and international markets. Our method uses a multi-head quantile neural network to predict a set of τ-quantiles based on stock characteristics and market variables. We interpolate these quantiles using cubic B-splines to derive the distribution and its moments, such as the mean, volatility, skewness, and kurtosis. This approach outperformed traditional parametric models and simpler neural networks in forecasting stock returns, providing more accurate out-of-sample mean and variance forecasts.

We also examined the pricing of distributional characteristics in the cross-section of stock returns using a comprehensive dataset of U.S. and international stocks. We found that only the central τ-quantiles are priced, with significant profitability of long-short portfolios based on these predictions. In addition, our results showed that the mean obtained by the numerical integration of the interpolated distribution leads to higher out-of-sample returns than the direct mean forecasts from other models. We found no evidence that other moments, such as variance, skewness, and kurtosis, are priced in the cross-section, adding to the ongoing debate in the asset pricing literature.

CRediT authorship contribution statement

Jozef Baruník: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Martin Hronec:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Ondřej Tobek:** Validation, Software, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2026.04.004.

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